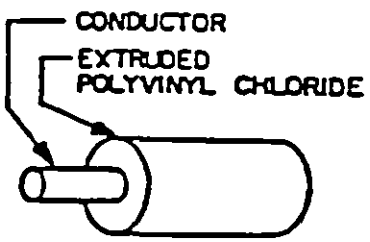
INCH-POUND
MIL-W-16878/2B(NAVY)
11 September 1992
SUPERSEDING
MIL-W-16878/2A(NAVY)
10 August 1981

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYVINYL CHLORIDE (PVC) INSULATED, 105°C, 1000 VOLTS (NOT FOR NAVY SHIPBOARD USE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-16878.



(NOT FOR NAVY SHIPBOARD USE)

FIGURE 1. Construction.

AMSC N/A FSC 6145

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TABLE I. Construction details.

(See footnotes at end of table.)

PIN 4/	Wire size	Strand-ing	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/ 3/	Coating		Min	Max
M16878/2-BDA*	26	1 X 26	Copper	Tin	0.0159	0.046	0.054
M16878/2-DDA*	26	1 X 26	H.S.C.A.	Silver	.0159	.046	.054
M16878/2-CDA*	26	1 X 26	C.C. steel	Tin	.0159	.046	.054
M16878/2-BDB*	26	7 X 34	Copper	Tin	.019	.049	.057
M16878/2-DDB*	26	7 X 34	H.S.C.A.	Silver	.019	.049	.057
M16878/2-BDE*	26	19 X 38	Copper	Tin	.021	.049	.057
M16878/2-DDE*	26	19 X 38	H.S.C.A.	Silver	.021	.049	.057
M16878/2-BEA*	24	1 X 24	Copper	Tin	.0201	.050	.058
M16878/2-DEA*	24	1 X 24	H.S.C.A.	Silver	.0201	.050	.058
M16878/2-CEA*	24	1 X 24	C.C. steel	Tin	.0201	.050	.058
M16878/2-BEB*	24	7 X 32	Copper	Tin	.024	.054	.062
M16878/2-DEB*	24	7 X 32	H.S.C.A.	Silver	.024	.054	.062
M16878/2-BEE*	24	19 X 36	Copper	Tin	.026	.054	.062
M16878/2-DEE*	24	19 X 36	H.S.C.A.	Silver	.026	.054	.062
M16878/2-BFA*	22	1 X 22	Copper	Tin	.0254	.055	.064
M16878/2-DFA*	22	1 X 22	H.S.C.A.	Silver	.0254	.055	.064
M16878/2-CFA*	1/ 22	1 X 22	C.C. steel	Tin	.0254	.055	.064
M16878/2-BFB*	22	7 X 30	Copper	Tin	.030	.060	.068
M16878/2-DFB*	22	7 X 30	H.S.C.A.	Silver	.030	.060	.068
M16878/2-BFE*	22	19 X 34	Copper	Tin	.033	.060	.068
M16878/2-DFE*	22	19 X 34	H.S.C.A.	Silver	.033	.060	.068
M16878/2-BGA*	20	1 X 20	Copper	Tin	.0320	.062	.070
M16878/2-DGA*	20	1 X 20	H.S.C.A.	Silver	.0320	.062	.070
M16878/2-BGB*	20	7 X 28	Copper	Tin	.038	.068	.076
M16878/2-DGB*	20	7 X 28	H.S.C.A.	Silver	.038	.068	.076
M16878/2-BGC*	20	10 X 30	Copper	Tin	.038	.068	.076
M16878/2-DGC*	20	10 X 30	H.S.C.A.	Silver	.038	.068	.076
M16878/2-BGE*	20	19 X 32	Copper	Tin	.041	.068	.076
M16878/2-DGE*	20	19 X 32	H.S.C.A.	Tin	.041	.068	.076
M16878/2-BHA*	18	1 X 18	Copper	Tin	.0403	.070	.079
M16878/2-BHB*	18	7 X 26	Copper	Tin	.049	.079	.087
M16878/2-BHD*	18	16 X 30	Copper	Tin	.047	.077	.085
M16878/2-BHE*	18	19 X 30	Copper	Tin	.051	.079	.087
M16878/2-BJA*	16	1 X 16	Copper	Tin	.0508	.080	.089
M16878/2-BJE*	16	19 X 29	Copper	Tin	.059	.089	.097
M16878/2-BJF*	16	26 X 30	Copper	Tin	.062	.092	.100
M16878/2-BKA*	14	1 X 14	Copper	Tin	.0641	.094	.102

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PIN 4/	Wire size	Strand-ing	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/ 3/	Coating		Min	Max
M16878/2-BKE*	14	19 X 27	Copper	Tin	0.072	0.102	0.110
M16878/2-BKH*	14	41 X 30	Copper	Tin	.080	.110	.118
M16878/2-BLA*	1/ 12	1 X 12	Copper	Tin	.0808	.110	.119
M16878/2-BLE*	12	19 X 25	Copper	Tin	.091	.121	.129
M16878/2-BLG*	12	37 X 28	Copper	Tin	.089	.119	.127
M16878/2-BLJ*	12	65 X 30	Copper	Tin	.098	.128	.136

1/ Inactive for new design.
2/ H.S.C.A. stands for high-strength copper alloy.
3/ C.C. stands for copper-clad.
4/ PIN stands for part or identifying number.

ADDITIONAL REQUIREMENTS:

Visual and mechanical examination: Required.
Spark test: 5.0 kV.
Impulse dielectric test: 10.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$.

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C.
K = 2000.
D = Maximum average diameter of finished wire.
d = Conductor diameter.

Cold bend: Condition 4 hours at minus 54 ± 1°C (see table II).

TABLE II. Cold bend mandrel sizes.

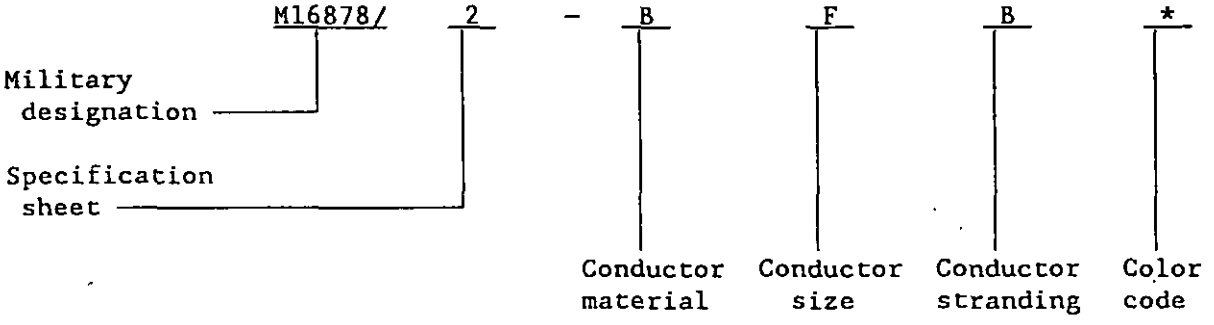
Wire size	Cold bend mandrel (maximum diameter in inches)
26 through 16	2
14, 12	3

Concentricity: 70 percent (minimum).
Surface resistance: Not required.
Wrap back: Not required.
Heat resistance: Condition at 150°C.
Flammability: Meets requirements of UL VW1 Flame Test.
Toxicity: Meets requirements of concentration levels specified in MIL-E-917.
Shrinkage: Required.

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Heating aging: 25 percent change (maximum) in 96 hours at 136°C.
Insulation tensile strength: 1800 pounds force per square inch (minimum).
Insulation elongation: 100 percent (minimum).
Marking and stripe durability: Required.
Fungus resistance: Required.
Maximum dc resistance of finished wire: See table I of MIL-W-16878.
Identification of product: Required for wire sizes 22 AWG and larger.

PIN example (see MIL-W-16878):



Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6145-N330-02)

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